

Q1 Sum the given number upto single digit

5 marks

Write a Python code to find the sum of the given number upto single digit

input: 89789

output : 5

Your Code

```
1 n=int(input())
2
3 while n >= 10:
4     s=0
5     while n > 0:
6         s += n % 10
7         n //= 10
8         n = s
9 print(n)
```

Input

89789

Output**Q2 Prime Number**

5 marks

Find the next prime number

input: 13

output: 17

Your Code

```
1 n=int(input())
2
3 n += 1
4
5 while True:
6     prime = True
7
8     for i in range(2,n):
9         if n % i == 0:
10             prime = False
11             break
12
13     if prime:
14         print(n)
15         break
16     n += 1
```

Input

13

Output

17

Q3 Swap the first and last digits

5 marks

WAP to swap the first and last digits of the given number.

Input: 23546

Output: 63542

Your Code

```
1 num=input()
2 result = num[-1] + num[1:-1] + num[0]
3 print(result)
```

Input

23546

Output

63542

Q4 Odd Sequence first and Even Sequence next

5 marks

WAP to odd Sequence first and Even Sequence next

Input: 658134

Output: 513684

Your Code

```
1 n=input()
2 result = n[1] + n[3] + n[4] + n[0] + n[2] + n[5]
3 print(result)
```

Input

658134

Output

513684

Your Code

```
1 n = input()
2 step = True
3 for i in range(len(n)-1):
4     if abs(int(n[i]) - int(n[i + 1])) != 1:
5         step = False
6         break
7
8 if step:
9     print("step number")
10 else:
11     print("not a step number")
```

Input

1235

Output

Q6 Reverse only the even numbers

5 marks

WAP to reverse only the even numbers

Input: 325649

Output: 345629

Your Code

```
1 n=input().strip()
2 evens = [ ch for ch in n if int(ch)%2 == 0]
3 result = []
4 for ch in n:
5     d = int(ch)
6     if d % 2 == 0:
7         result.append(evens.pop())
8     else:
9         result.append(ch)
10 print("".join(result))
```

Input

325649

Output

345629

Q7 Encoding

5 marks

WAP to encode the given number

Input: 55553322255

Output: 54322352

Your Code

```
1 n = input().strip()
2 result = ""
3 count = 1
4 for i in range(1, len(n)):
5     if n[i] == n[i-1]:
6         count += 1
7     else:
8         result += n[i-1] + str(count)
9         count = 1
10 result += n[-1] + str(count)
11 print(result)
```

Input

55553322255

Output

54322352

Q8 Reverse Odd Digit

5 marks

WAP to Reverse Odd Digits Only

Input: 816253

Output: 351

Your Code

```
1 n = input().strip()
2 odd = ""
3 for ch in n:
4     if int(ch) % 2 != 0:
5         odd += ch
6 print(odd[::-1])
```

Input

816253

Output

351

Q9 Convert every number into the next sequence number.

5 marks

WAP to convert every number into the next sequence number.

Input: 89781

Output: 90892

Your Code

```
1 n = input().strip()
2 result = ""
3 for ch in n:
4     if ch == '9':
5         result += '0'
6     else:
7         result += str(int(ch) + 1)
8 print(result)
```

Input

89781

Output

90892

Q10 Print the total factor count of a given number

5 marks

WAP to print the total factor count of a given number

Input: 12

Output: 6

Your Code

```
1 n = int(input())
2 count = 0
3 for i in range(1, n+1):
4     if n % i == 0:
5         count += 1
6 print(count)
```

Input

12

Output

6